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Summary Lecture 19:
The pseudoinverse of a matrix,
see Chapter 6.4

The Pseudoinverse or Moore–Penrose Inverse

Our next task

construct an analogue to the inverse of a matrix A for matrices that have no inverse. This is called the pseudoinverse and we will denote it by $A^\dagger$.

The hurdles to overcome

- For some vectors b there might not be a vector x such that $Ax = b$.
- For some vectors b there may be more than one x such that $Ax = b$ and we must pick one.
- Even if we make such choices, it is not clear that such operation will correspond to multiplying by a matrix $A^\dagger$.

Our plan to take the hurdles

- Develop a pseudoinverse for matrices with full column rank.
- Develop a pseudoinverse for matrices with full row rank.
- Write a general matrix as as product of two matrices: one of full column rank and one of full row rank.

Pseudoinverse for matrices with full column rank

The intuition

If the columns of A are linearly independent it makes sense to build $A^\dagger$ such that $A^\dagger b$ is the Least Squares Solution $\hat{x} = (A^\top A)^{-1} A^\top b$ (the vector $\hat{x}$ such that $A\hat{x}$ is as close as possible to b).

Definition (Pseudoinverse for matrices with full column rank)

For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = n$ we define the pseudo-inverse $A^\dagger \in \mathbb{R}^{n \times m}$ of A as

$$A^\dagger = (A^\top A)^{-1} A^\top.$$

Proposition

For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = n$, the pseudoinverse $A^\dagger$ is a left inverse of A , meaning that $A^\dagger A = I$.

Pseudoinverse for matrices with full row rank

The intuition: If the rows of A are linearly independent, then A^T has full column rank and we use the pseudo-inverse of A^T to define a pseudo-inverse of A .

Definition (Pseudoinverse for matrices with full row rank)

For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$ the pseudo-inverse of $A^\dagger \in \mathbb{R}^{n \times m}$ of A is

$$A^\dagger = A^T (AA^T)^{-1}.$$

Properties: Lemmas 6.4.4, 6.4.5, Proposition 6.4.6

- For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$, $AA^\dagger = I$.
- $\hat{x} = A^\dagger b$ is the solution to

$$\min\{\|x\|^2 \mid x \in \mathbb{R}^n, Ax = b\}.$$

- $\hat{x} = A^\dagger b \in C(A^T)$.

Pseudoinverse in general

Use the CR decomposition $A = CR$: C full column rank, R full row rank.

Definition (Pseudoinverse for all matrices)

$A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = r$ with CR decomposition $A = CR$. Let $A^\dagger = R^\dagger C^\dagger$.

Lemma 6.4.8

For $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^n$, the (unique) solution to $(*)$ is given by $\hat{x} = A^\dagger b$.

$$(*) \quad \min \left\{ \|x\|^2 \text{ s.t. } x \in \mathbb{R}^n, A^\top A x = A^\top b \right\}$$

Theorem 6.4.11

- ① $AA^\dagger A = A$ and $A^\dagger AA^\dagger = A^\dagger$.
- ② $AA^\dagger$ is symmetric. It is the projection matrix for projection on $C(A)$,
- ③ $A^\dagger A$ is symmetric. It is the projection matrix for projection on $C(A^\top)$.
- ④ $(A^\top)^\dagger = (A^\dagger)^\top$.